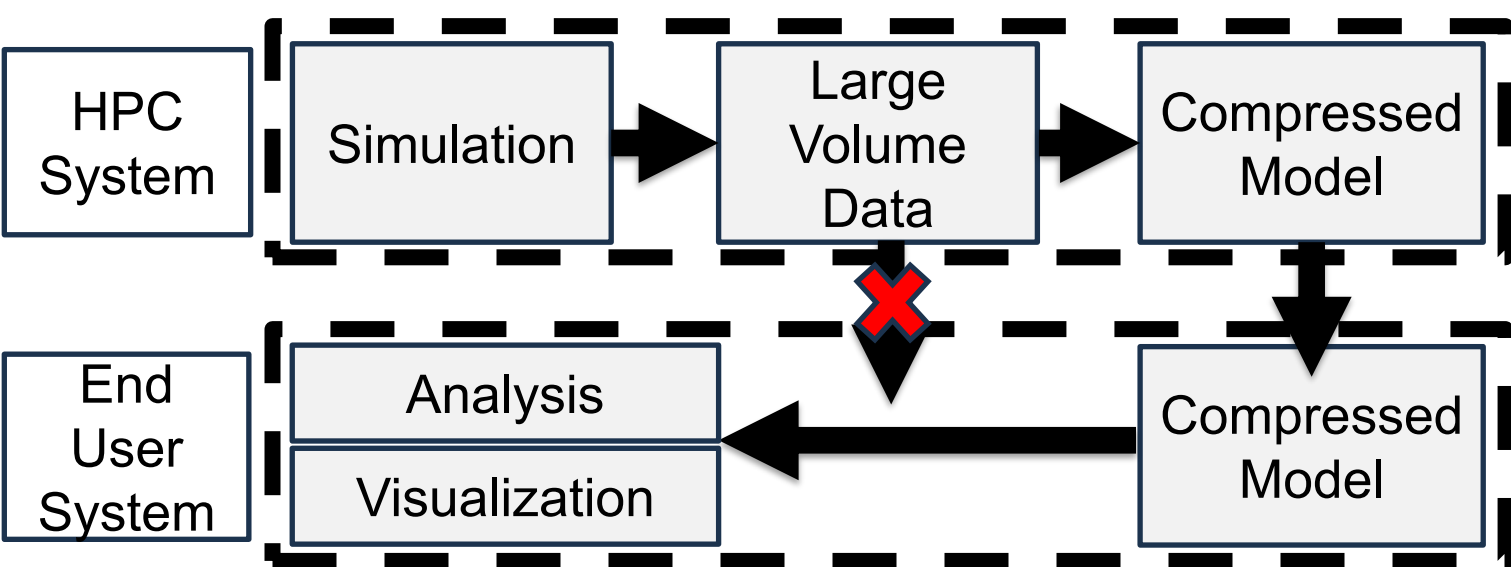


Introduction

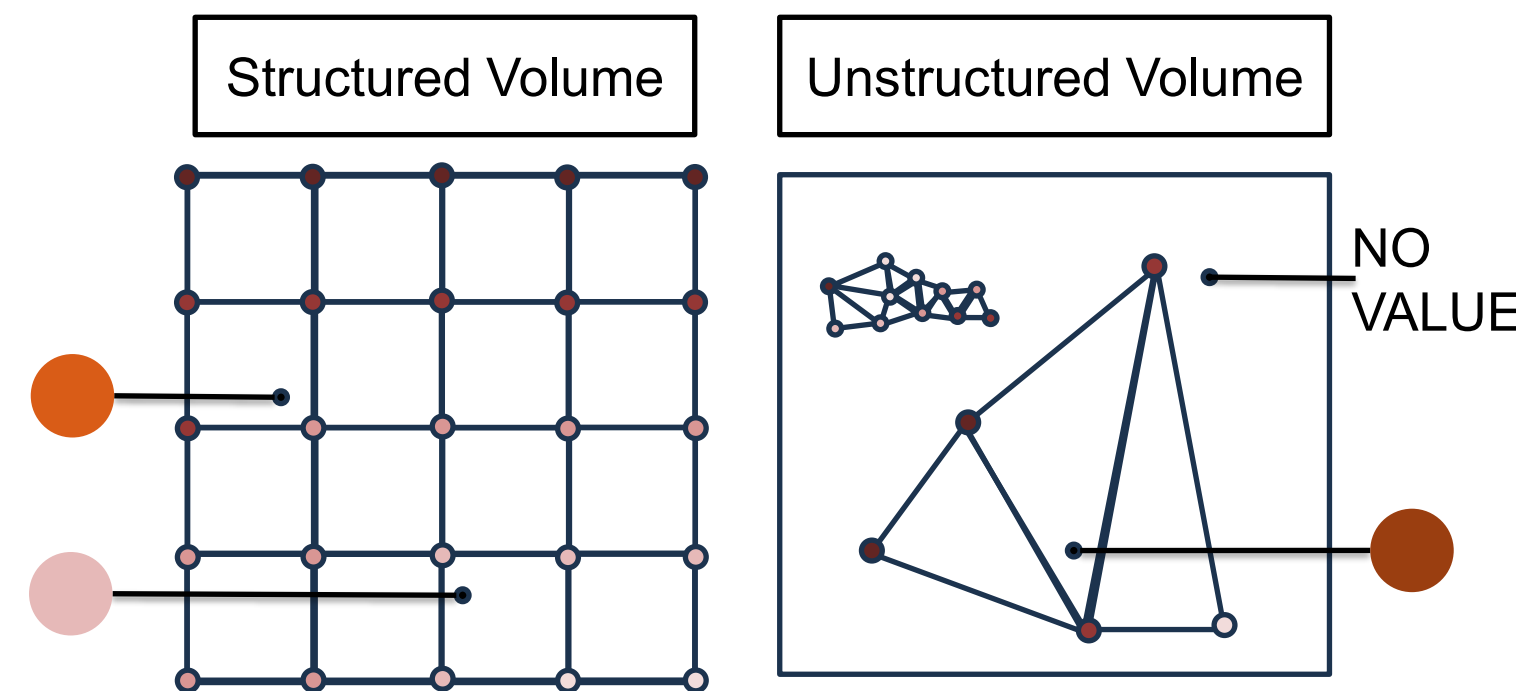
Why are new compression methods needed?

Effectively using large-scale volume datasets is a challenge, as they exceed the memory of user systems. To address this, methods are needed that can reduce memory footprints and maintain interactivity. Traditional methods achieve high compression ratios, but require decompression before data can be used, prohibiting or greatly slowing end user tasks.



Where does existing work fail?

A popular family of methods for this use case are scene representation networks (SRNs), which approximate volume data with neural networks that learn a function from input positions to output values. However, these networks do not encode unstructured volume geometry. As a result, the volume's surface must be stored for later tasks, setting an extremely limiting hard cap on compression ratios.



Evaluation

We evaluate our method against state-of-the-art SRNs for structured and unstructured volumes, trained on an NVIDIA A100 GPU. We measure reconstruction quality (PSNR) and training time (seconds) across different model compression ratios (dataset size / model size). For unstructured data, we also include "Real CR," which adds needed geometry to model size.

Structured Datasets	Vertebra	Miranda	Chameleon	Richtmyer	Unstructured Datasets	Mito	SF1	Earthquake	Impact
	537 MB	4.29 GB	4.53 GB	32.21 GB		246 MB	622 MB	4.31 GB	17.93 GB

Unstructured Volume Evaluation

Dataset	Model CR	UGINR			Ours		
		PSNR	Time	Real CR	PSNR	Time	Real CR
Mito	64x	37.84	31	16.6x	58.68	23	64x
	256x	28.80	31	20.5x	55.88	20	256x
	1024x	23.15	33	22.0x	52.58	19	1024x
SF1	64x	42.73	73	17.3x	53.14	29	64x
	256x	35.55	75	21.7x	50.20	26	256x
	1024x	27.05	74	23.2x	45.45	25	1024x
Earthquake	64x	51.98	1456	6.9x	54.82	146	64x
	256x	50.61	1435	7.5x	50.85	71	256x
	1024x	45.97	1488	7.6x	44.11	23	1024x
Impact	64x	OOM	OOM	OOM	48.33	446	64x
	256x	OOM	OOM	OOM	46.86	135	256x
	1024x	OOM	OOM	OOM	44.77	47	1024x

Our model achieves on average **1.43x** higher quality and **4.53x** faster training times while reaching real compression ratios up to **135x** higher than previously possible

Future Work

Future work will be on:

- Extending to multivariate and time-varying data
- Implementing Gaussian model compression with quantization
- Designing rendering algorithms for efficiently raytracing our models

Acknowledgement



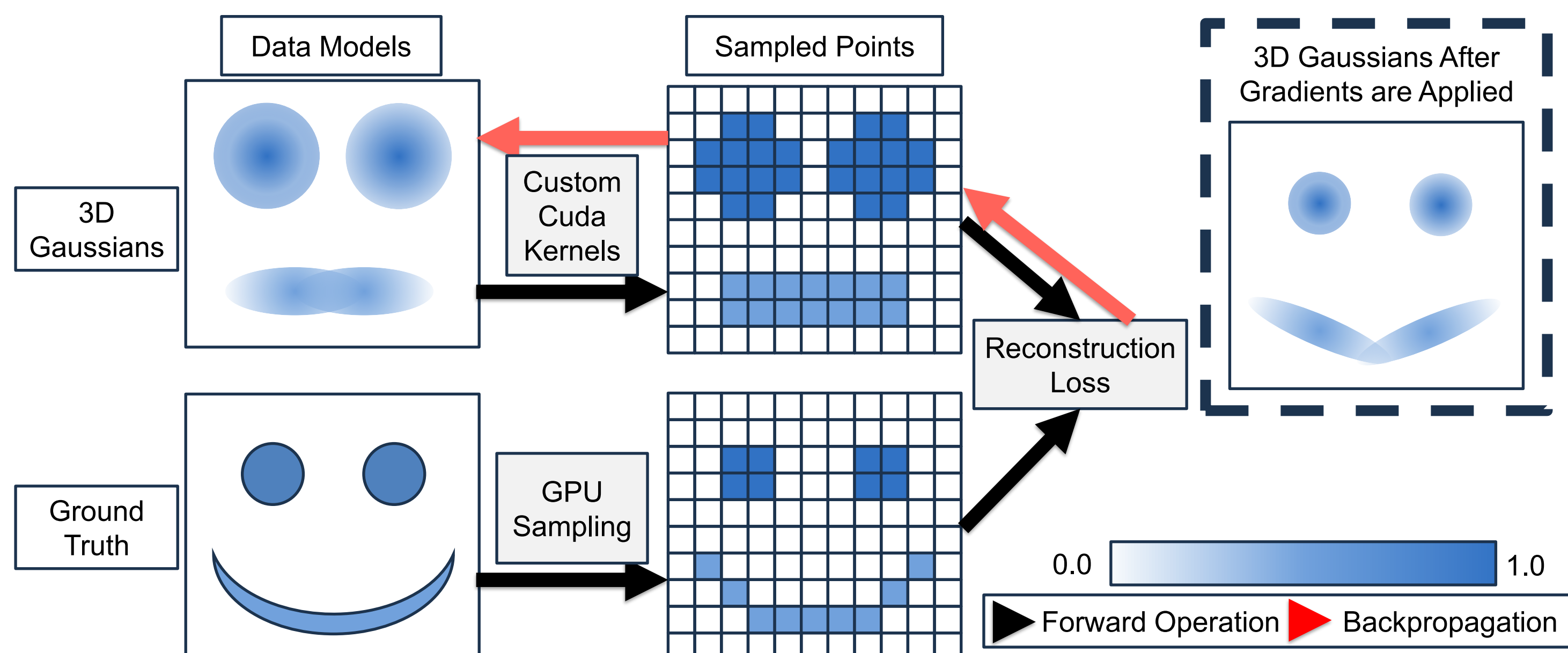
Our Method

What are 3D Gaussians?

In this work, we explore compressing scientific volumes by approximating them as collections of 3-dimensional Gaussians. Each Gaussian is defined as a scaled and rotated distribution around a point in 3D space, and output is computed through averaging the values of Gaussians intersecting any location. By using fewer Gaussians than data points in the ground truth, these models represent a dataset using a fraction of the original memory. Because the Gaussians are explicitly defined in space, they naturally encode geometry for unstructured volumes.

How are 3D Gaussian models trained?

Models are trained to reconstruct a volume using gradient descent. This is done by sampling the model and the ground truth, computing the loss between the two sets of sampled points, then backpropagating gradients to each Gaussian.



Contributions

Our contributions:

- The first *explicit* learned model for volume data compression, removing the need to store mesh geometry for unstructured volumes. At the same model sizes, our method achieves on average **1.43x** higher quality and **4.53x** faster training times than the SotA while reaching compression ratios up to **135x** higher than previously possible.
- A novel sampling-error based training pipeline for Gaussian models. This method outperforms even the fastest SRN on training time by an average **1.52x**.

Structured Volume Evaluation

Dataset	Model CR	IVNR 20k		IVNR 200k		AMGSRN		Ours	
		PSNR	Time	PSNR	Time	PSNR	Time	PSNR	Time
Vertebra	1024x	37.29	<u>10</u>	<u>37.56</u>	101	38.23	120	37.44	2
	256x	39.20	<u>11</u>	39.65	105	<u>39.45</u>	108	39.19	4
	64x	40.98	<u>15</u>	41.52	153	<u>41.92</u>	120	41.98	6
Miranda	1024x	31.63	<u>12</u>	32.73	116	<u>32.20</u>	149	31.45	7
	256x	<u>37.77</u>	<u>17</u>	40.76	166	36.42	221	37.36	11
	64x	39.16	<u>36</u>	43.22	355	42.71	213	<u>42.98</u>	35
Chameleon	1024x	41.78	<u>11</u>	42.98	112	<u>43.65</u>	133	44.01	6
	256x	46.84	<u>16</u>	48.40	157	46.29	180	<u>47.86</u>	15
	64x	48.02	31	50.50	310	47.76	176	<u>49.74</u>	<u>49</u>
Richtmyer	1024x	22.65	<u>19</u>	23.40	188	27.38	840	<u>26.53</u>	9
	256x	23.67	48	25.45	454	29.38	863	<u>28.61</u>	<u>51</u>
	64x	24.10	90	26.91	859	31.12	896	<u>29.83</u>	<u>140</u>

Our model outperforms the fastest structured data SRN on training time by an average **1.52x**, while reaching reconstruction quality equal to much slower models.

